Reproductive_senescence_BaFTA

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Setup

Load packages

```
library(here)
library(tidyverse)
library(cowplot)
library(ggplot2)
library(gridExtra)
library(snowfall)
library(BaFTA)
library(flextable)
library(officer)
library(lme4)
library(lmerTest)
library(emmeans)
```

Plot style

```
theme_publication <- theme_classic() +
  theme(
    axis.text  = element_text(size = 11),
    axis.title = element_text(size = 12),
    legend.text  = element_text(size = 11),
    legend.title = element_text(size = 12)
  )
```

Load and prepare data

```
# Load data
data <- read.csv(here("data", "processed", "reproductive_data.csv"),
                 na = c("", "NA", ".", "?"))[, c('Year', 'FemID', 'Site', 'Age', 'Clutch')]

# Rename columns
```

```r
data <- data %>%
  rename(
    year = Year,
    indID = FemID,
    site = Site,
    nOffspring = Clutch)

# Format variables
data$indID <- as.factor(data$indID)
data$site <- as.factor(data$site)
data$Age <- as.integer(data$Age)
data$nOffspring <- as.integer(data$nOffspring)

# Remove missing data
data <- subset(data,
              !is.na(indID) & indID != 0 &
              !is.na(Age) & Age != 0 &
              !is.na(nOffspring) & nOffspring != 0)
# NOTE: Zero litter sizes are excluded because non-gravid status, stillbirths,
# and mid-pregnancy reabsorptions were not consistently recorded throughout the
# study. Zeros therefore cannot be distinguished from missing data, so this
# analysis models conditional fertility (litter size given reproduction occurred).

head(data)
```

```
##      year indID site Age nOffspring
## 1354 2002   205   OR   2          2
## 1524 2002   247   OR   2          1
## 1908 2003   304   OR   2          1
## 1967 2003   315   OR   3          2
## 1981 2003   317   OR   3          2
## 2066 2003   338   OR   3          2
```

```r
cat("Final n:", nrow(data), "\n")
```

```
## Final n: 1898
```

Subset by population

```r
dataCP <- subset(data, site == "CP")
dataOR <- subset(data, site == "OR")

cat("CP sample size:", nrow(dataCP), "\n")
```

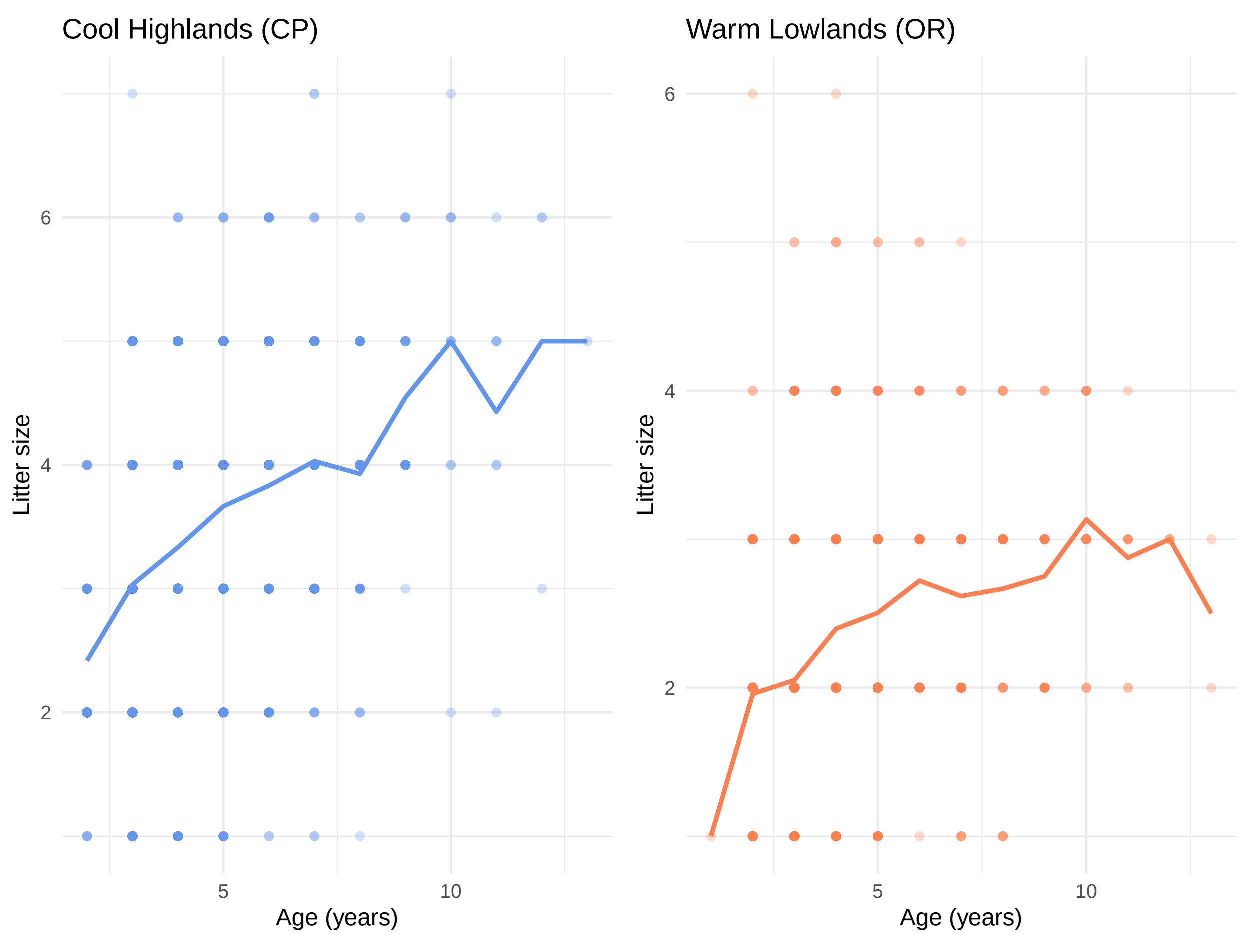
```
## CP sample size: 1181
```

```r
cat("OR sample size:", nrow(dataOR), "\n")
```

```
## OR sample size: 717
```

Visualize raw data

```
# CP population
p_raw_CP <- ggplot(dataCP, aes(x = Age, y = nOffspring)) +
  geom_point(alpha = 0.3, color = "cornflowerblue") +
  stat_summary(fun = mean, geom = "line", color = "cornflowerblue", size = 1) +
  labs(title = "Cool Highlands (CP)", x = "Age (years)", y = "Litter size") +
  theme_minimal()

# OR population
p_raw_OR <- ggplot(dataOR, aes(x = Age, y = nOffspring)) +
  geom_point(alpha = 0.3, color = "coral") +
  stat_summary(fun = mean, geom = "line", color = "coral", size = 1) +
  labs(title = "Warm Lowlands (OR)", x = "Age (years)", y = "Litter size") +
  theme_minimal()

# Display
plot_grid(p_raw_CP, p_raw_OR, ncol = 2)
```



Pattern: Both populations show early increase. Orford plateaus around 5. CP continues increasing and maybe plateaus at 10.

Model Fitting - CP Population

CP: Quadratic model

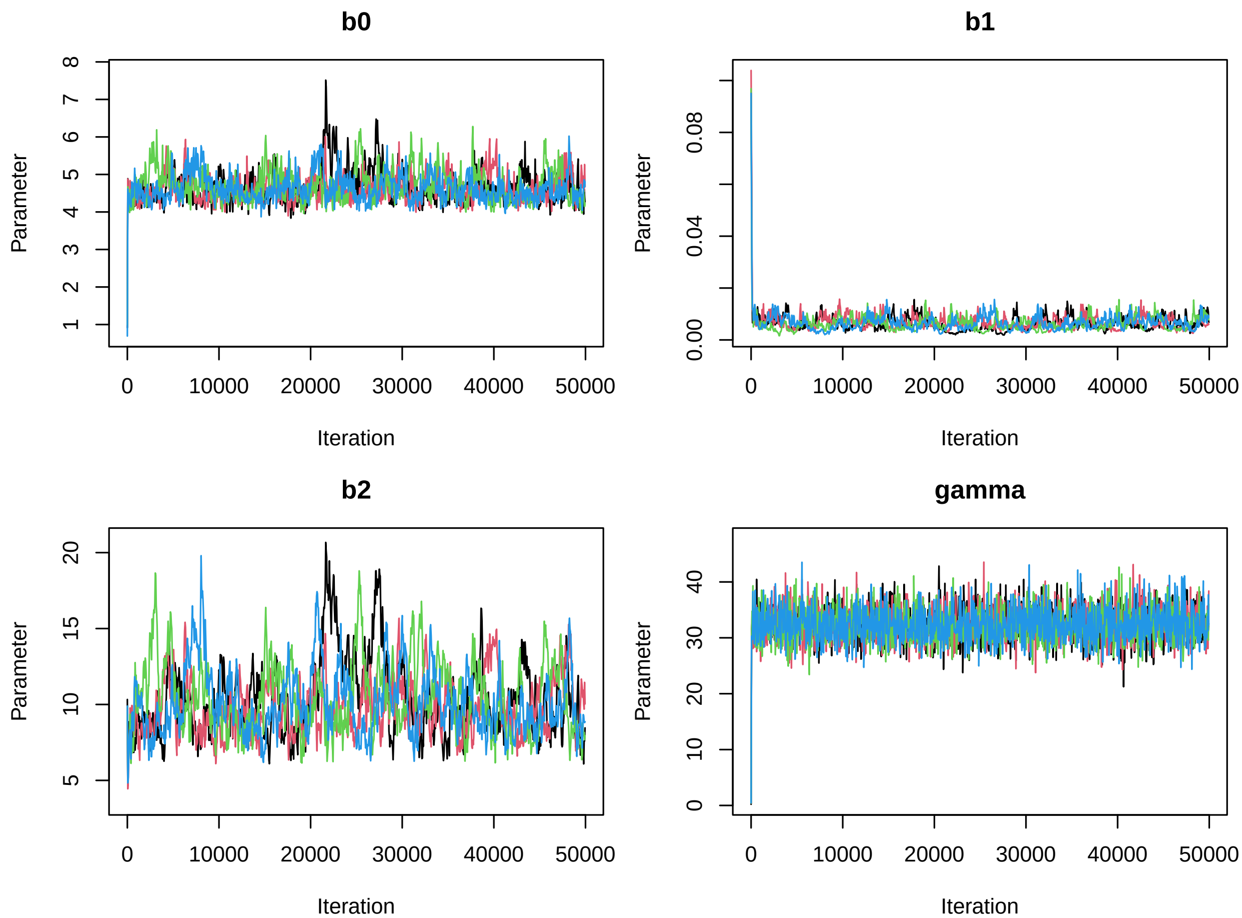
```r
# Run model (long computation)
set.seed(123)

out_quad_CP <- bafta(object = dataCP, model = "quadratic",
                     dataType = "indivSimple", nsim = 4, ncpus = 4,
                     niter = 50000, burnin = 10000, thinning = 50)

# Save output
dir.create(here("bafta_outputs"), showWarnings = FALSE)
saveRDS(out_quad_CP, here("bafta_outputs", "out_quad_CP.rds"))
```

```r
# Load saved model
out_quad_CP <- readRDS(here("bafta_outputs", "out_quad_CP.rds"))
out_quad_CP
```

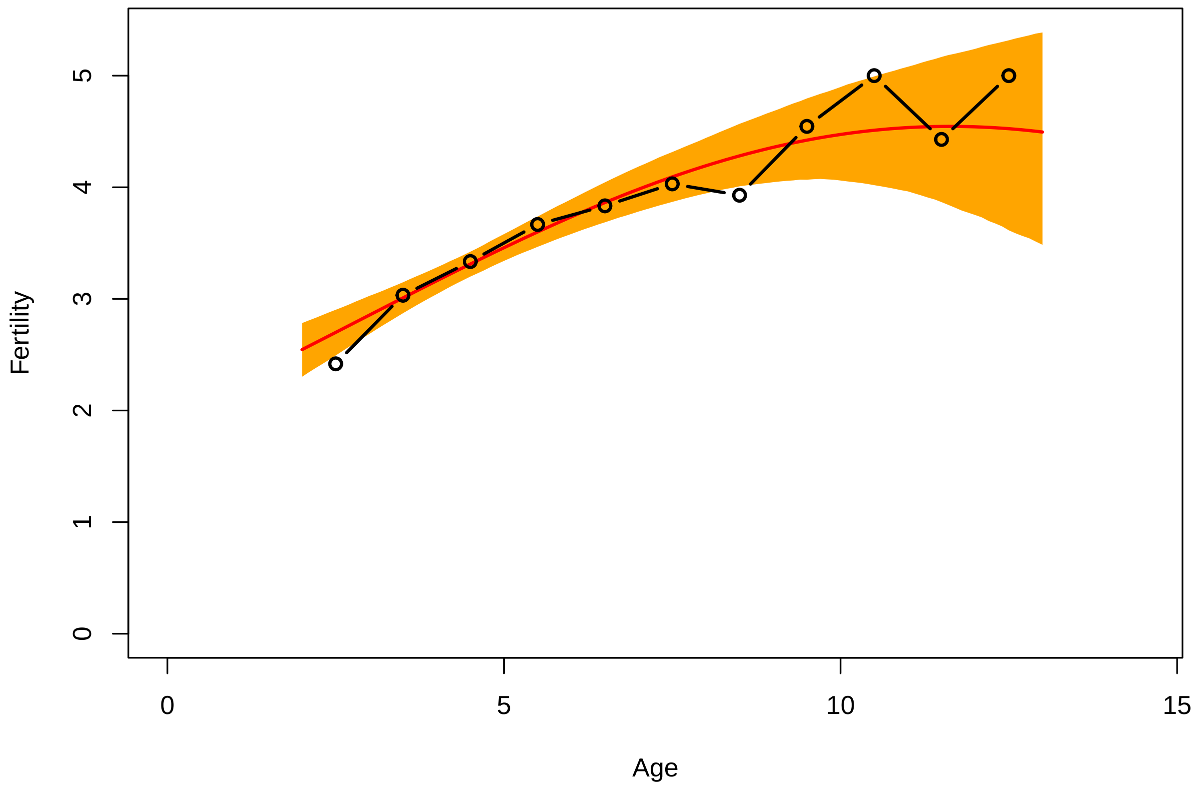
```
##
## Call:
## Model             : quadratic
## Data type         : indivSimple
## Num. iterations   : 50000
## Burnin            : 10000
## Thinning          : 50
## Number of sims.   : 4
## Computing time    : 2.22 mins
##
## Coefficients:
##            Mean       SD     Lower    Upper Neff  Rhat
## b0     4.663095 0.398021  4.099779  5.62033  692 1.007
## b1     0.006406 0.002368  0.002884  0.01176   67 1.001
## b2    10.198486 2.212958  6.994347 15.44799  327 1.010
## gamma 32.317408 2.961111 26.765096 38.32012 2713 1.000
##
## Convergence:
## All parameter chains converged.
##
## Model fit:
##
## DIC = 4128.85
##
## Predictive loss:
##  Good. Fit Penalty Deviance
##       1312    4465     5777
```

Results: - Convergence: All parameter chains converged - DIC = 4128.85 - Predictive loss: Good (Fit = 1312, Penalty = 4465, Deviance = 5777) - Smallest Neff = 67 Critically low (should be over 400) - parameter estimates unreliable

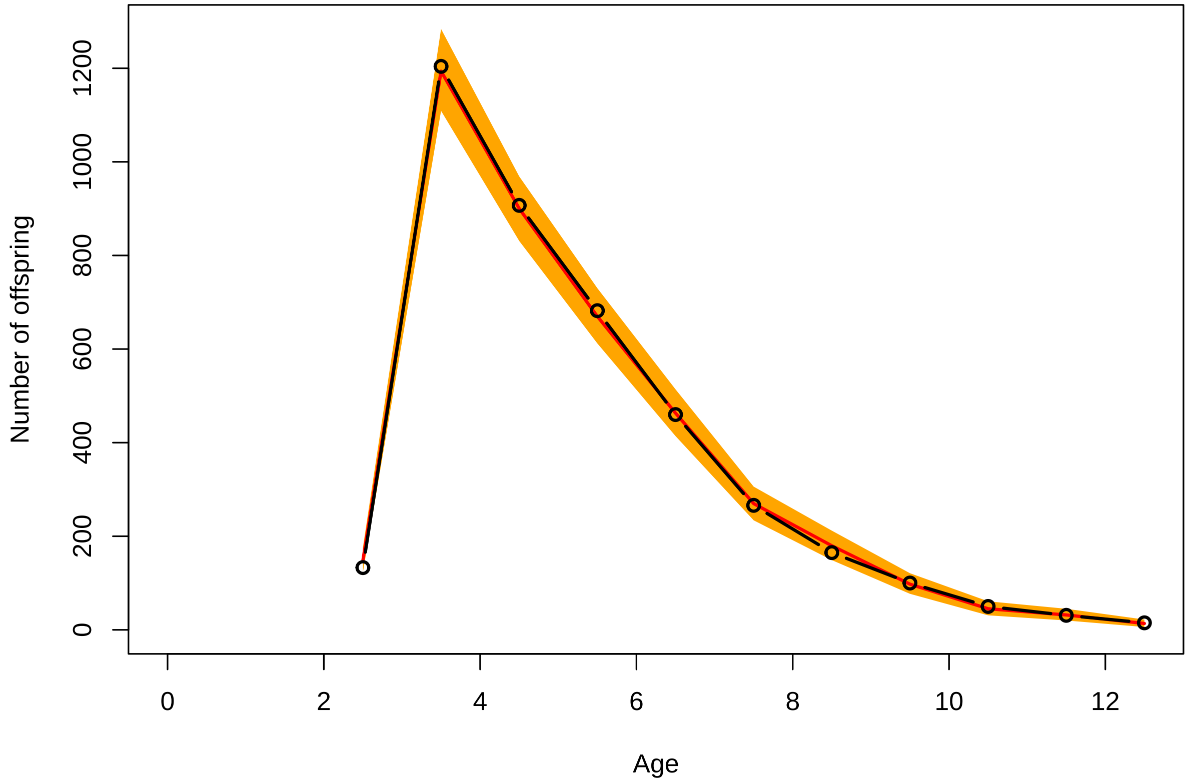
```r
# Diagnostic plots
plot(out_quad_CP)
```




```r
plot(out_quad_CP, type = "fertility")
```




```r
plot(out_quad_CP, type = "predictive")
```



CP: Gamma model

```r
# Run model (long computation)
set.seed(123)

out_gamma_CP <- bafta(object = dataCP, model = "gamma",
                      dataType = "indivSimple", nsim = 4, ncpus = 4,
                      niter = 50000, burnin = 10000, thinning = 50)

# Save output
saveRDS(out_gamma_CP, here("bafta_outputs", "out_gamma_CP.rds"))
```

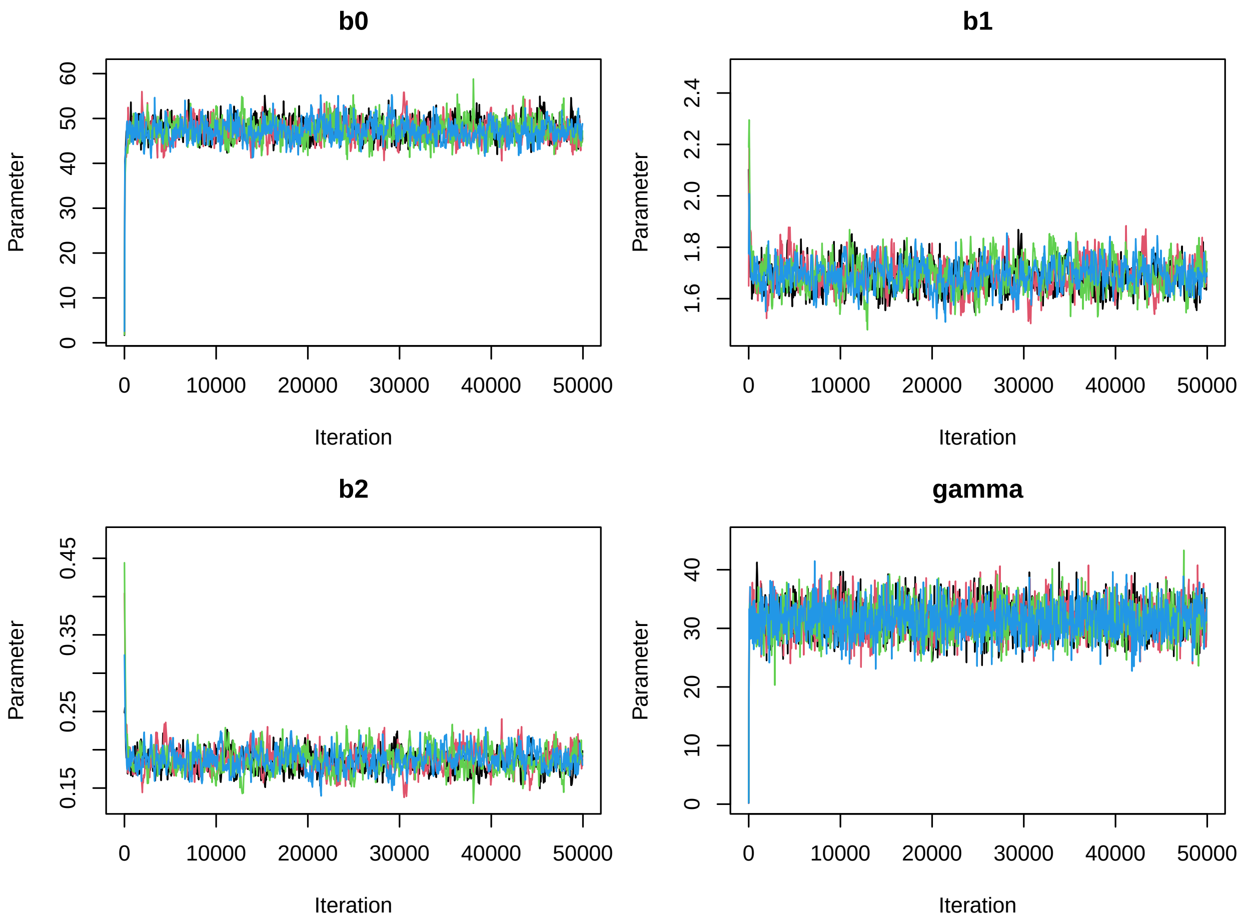
```r
# Load saved model
out_gamma_CP <- readRDS(here("bafta_outputs", "out_gamma_CP.rds"))
out_gamma_CP
```

```
##
## Call:
## Model              : gamma
## Data type          : indivSimple
## Num. iterations    : 50000
```

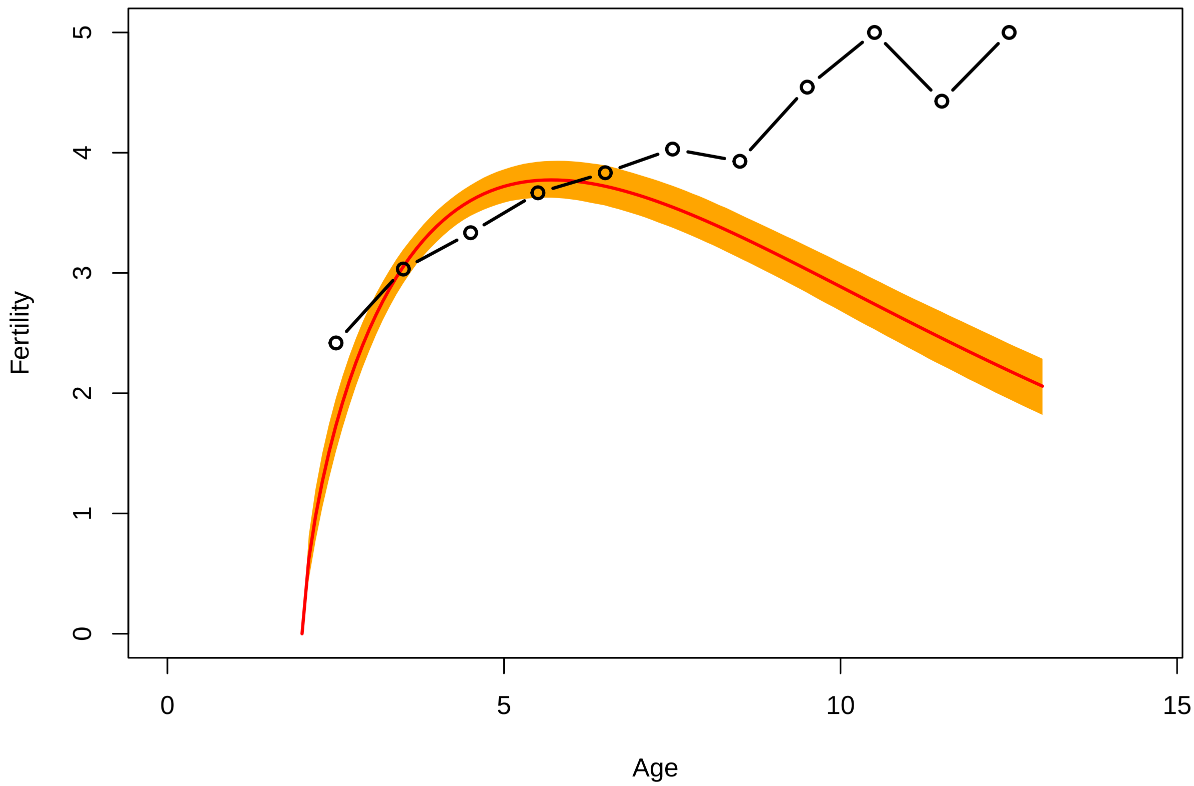
```
## Burnin           : 10000
## Thinning         : 50
## Number of sims.  : 4
## Computing time   : 4.64 mins
##
## Coefficients:
##          Mean      SD   Lower   Upper Neff  Rhat
## b0    47.5142 2.35097 43.0133 52.2267 1011 1.003
## b1     1.6897 0.05783  1.5795  1.8106  980 1.005
## b2     0.1861 0.01488  0.1584  0.2169  791 1.005
## gamma 31.5163 2.88458 26.1218 37.4754 2687 1.000
##
## Convergence:
## All parameter chains converged.
##
## Model fit:
##
## DIC = 4237.84
##
## Predictive loss:
##  Good. Fit Penalty Deviance
##       1540    4358     5899
```

Results: - Convergence: All parameter chains converged - DIC = 4237.84 - Predictive loss: Good (Fit = 1540, Penalty = 4358, Deviance = 5899) - Smallest Neff = 791 - OK

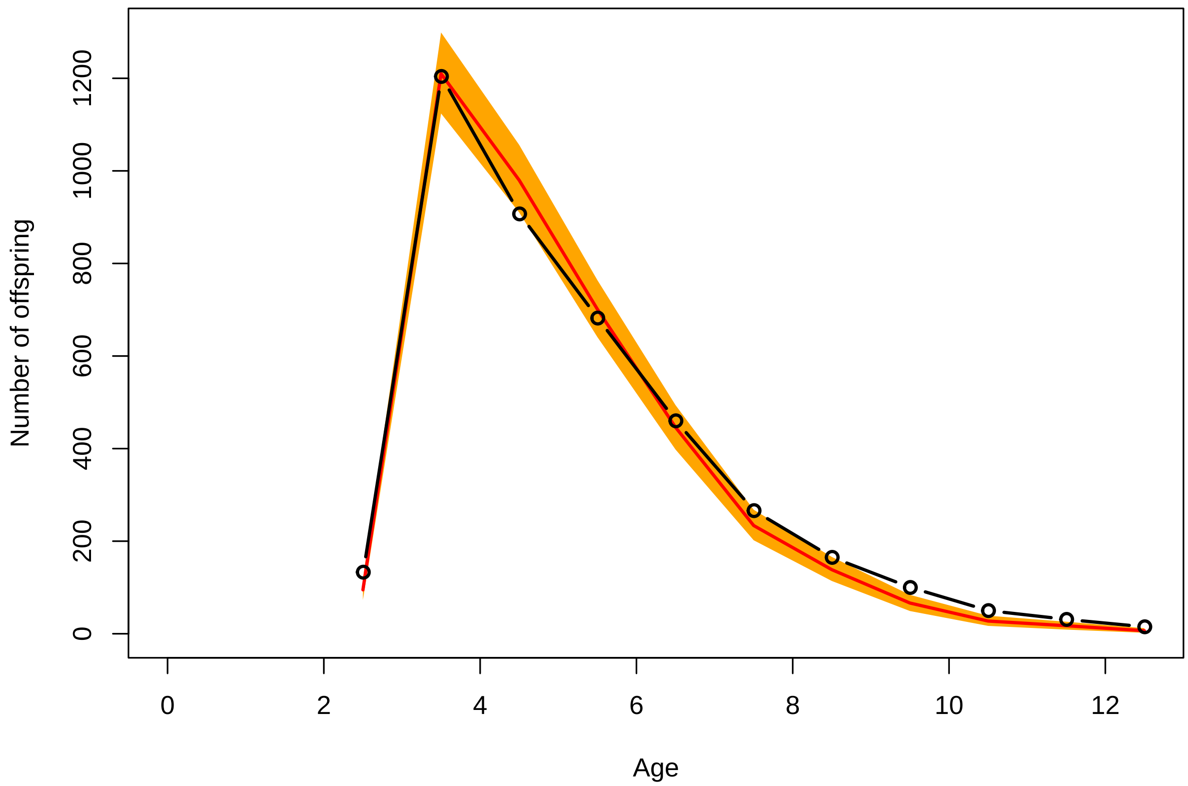
```r
# Diagnostic plots
plot(out_gamma_CP)
```




```
plot(out_gamma_CP, type = "fertility") #not good
```




```r
plot(out_gamma_CP, type = "predictive")
```



CP: Colchero-Muller model

```
# Run model (long computation)
set.seed(123)

out_CM_CP <- bafta(object = dataCP, model = "ColcheroMuller",
                   dataType = "indivSimple", nsim = 4, ncpus = 4,
                   niter = 50000, burnin = 10000, thinning = 50)

# Save output
saveRDS(out_CM_CP, here("bafta_outputs", "out_CM_CP.rds"))
```

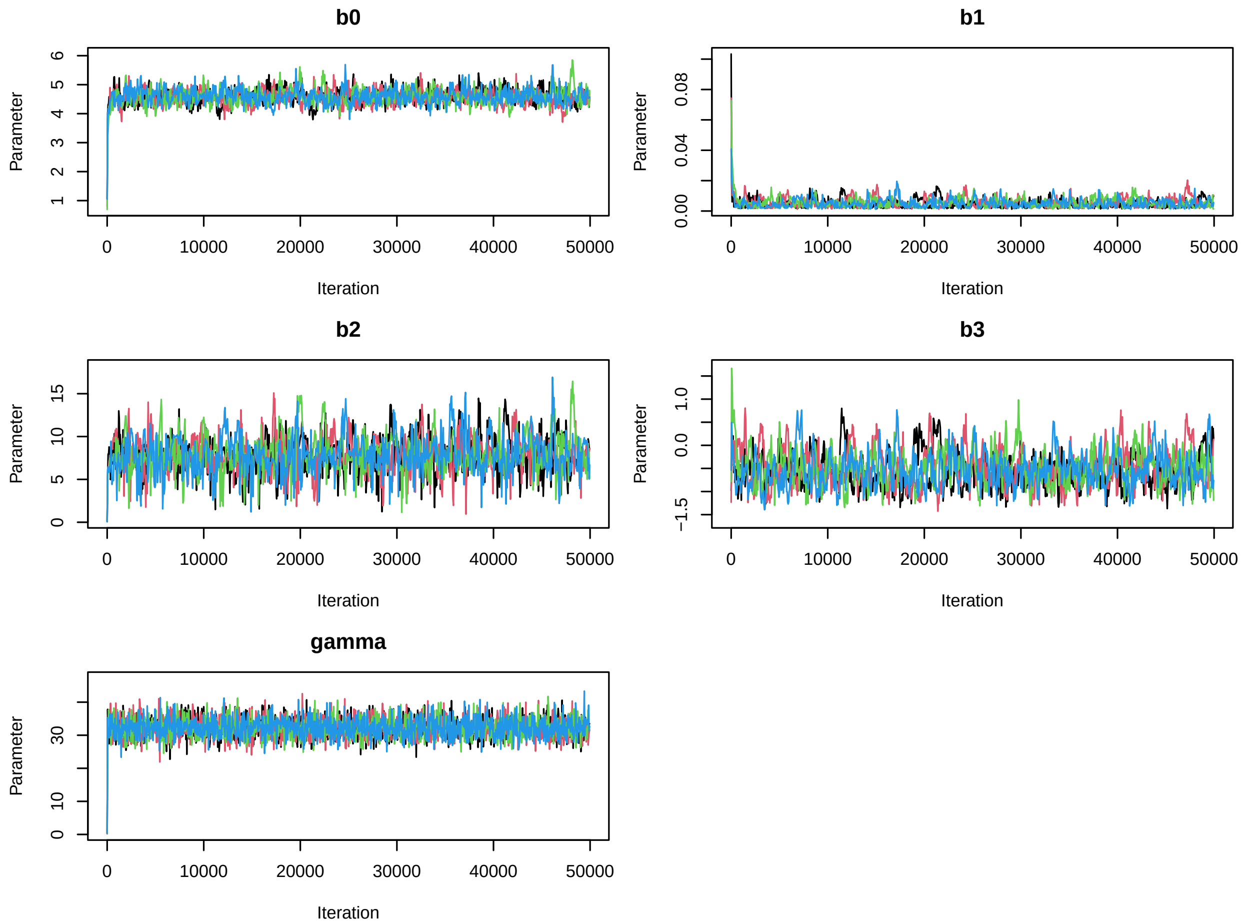
```
# Load saved model
out_CM_CP <- readRDS(here("bafta_outputs", "out_CM_CP.rds"))
out_CM_CP
```

```
##
## Call:
## Model              : ColcheroMuller
## Data type          : indivSimple
## Num. iterations    : 50000
```

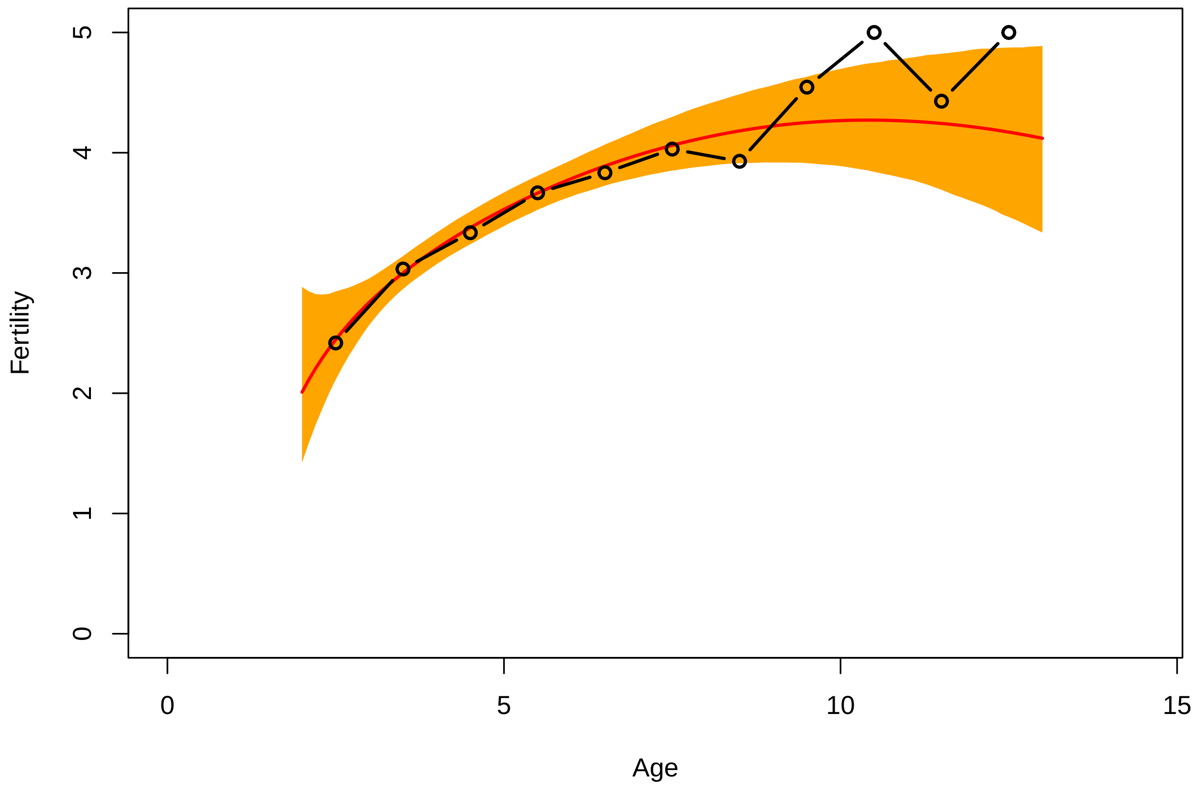
```
## Burnin           : 10000
## Thinning         : 50
## Number of sims.  : 4
## Computing time   : 3.86 mins
##
## Coefficients:
##            Mean       SD     Lower    Upper Neff  Rhat
## b0     4.588669 0.257074  4.116747  5.11211 1007 1.005
## b1     0.005144 0.002912  0.001662  0.01235  614 1.005
## b2     7.916850 2.106235  3.664397 12.55054  951 1.002
## b3    -0.513591 0.374995 -1.153916  0.31766  655 1.005
## gamma 32.349824 2.900753 27.183494 38.53556 2738 1.000
##
## Convergence:
## All parameter chains converged.
##
## Model fit:
##
## DIC = 4128.57
##
## Predictive loss:
##  Good. Fit Penalty Deviance
##       1307    4465     5772
```

Results: - Convergence: All parameter chains converged - DIC = 4128.57 - Predictive loss: Good (Fit = 1307, Penalty = 4465, Deviance = 5772) - Smallest Neff = 614 - OK

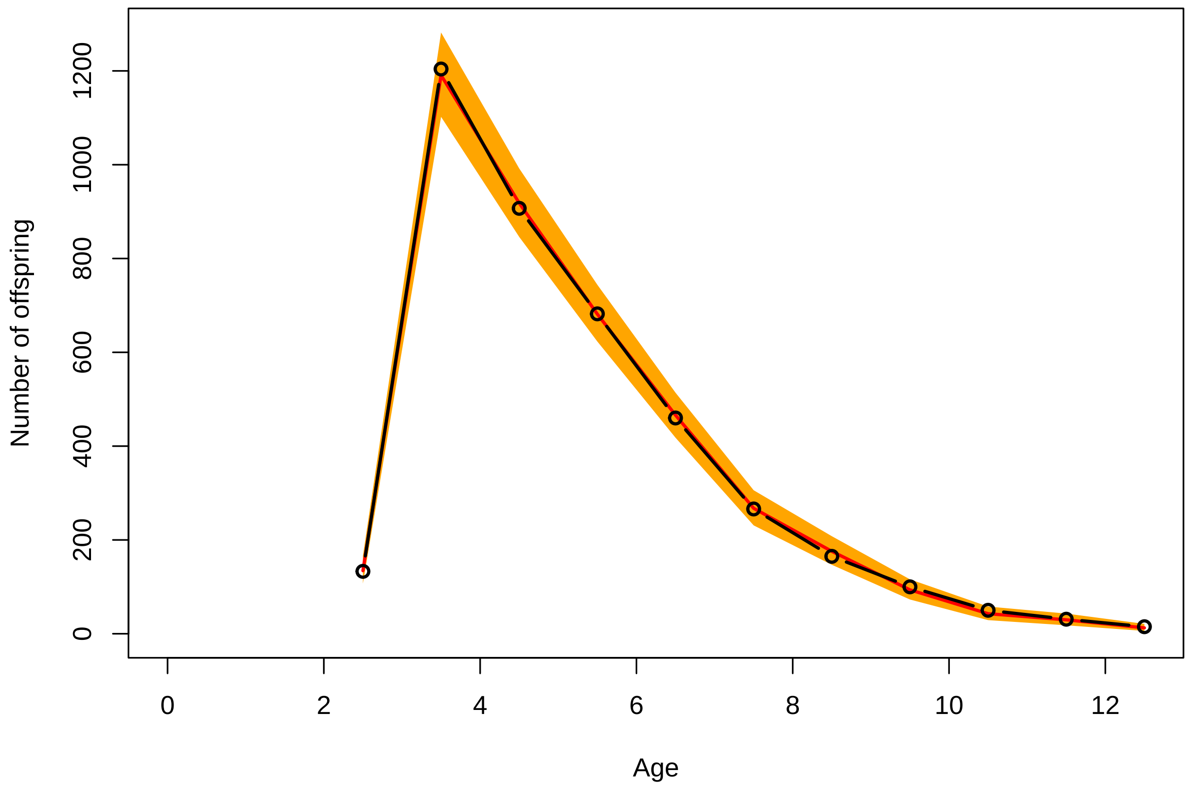
```r
# Diagnostic plots
plot(out_CM_CP)
```




```r
plot(out_CM_CP, type = "fertility")
```




```r
plot(out_CM_CP, type = "predictive")
```



CP Model comparison

Best model for CP: Colchero-Muller (DIC = 4128.57) - Essentially the same DIC as quadratic (which had very low neff for b2) - Visually, both quadratic and CM fit data well - CM model preferred due to better convergence

---

Model Fitting - OR Population

OR: Quadratic model

```
# Run model (long computation)
set.seed(123)

out_quad_OR <- bafta(object = dataOR, model = "quadratic",
                     dataType = "indivSimple", nsim = 4, ncpus = 4,
                     niter = 50000, burnin = 10000, thinning = 50)
```

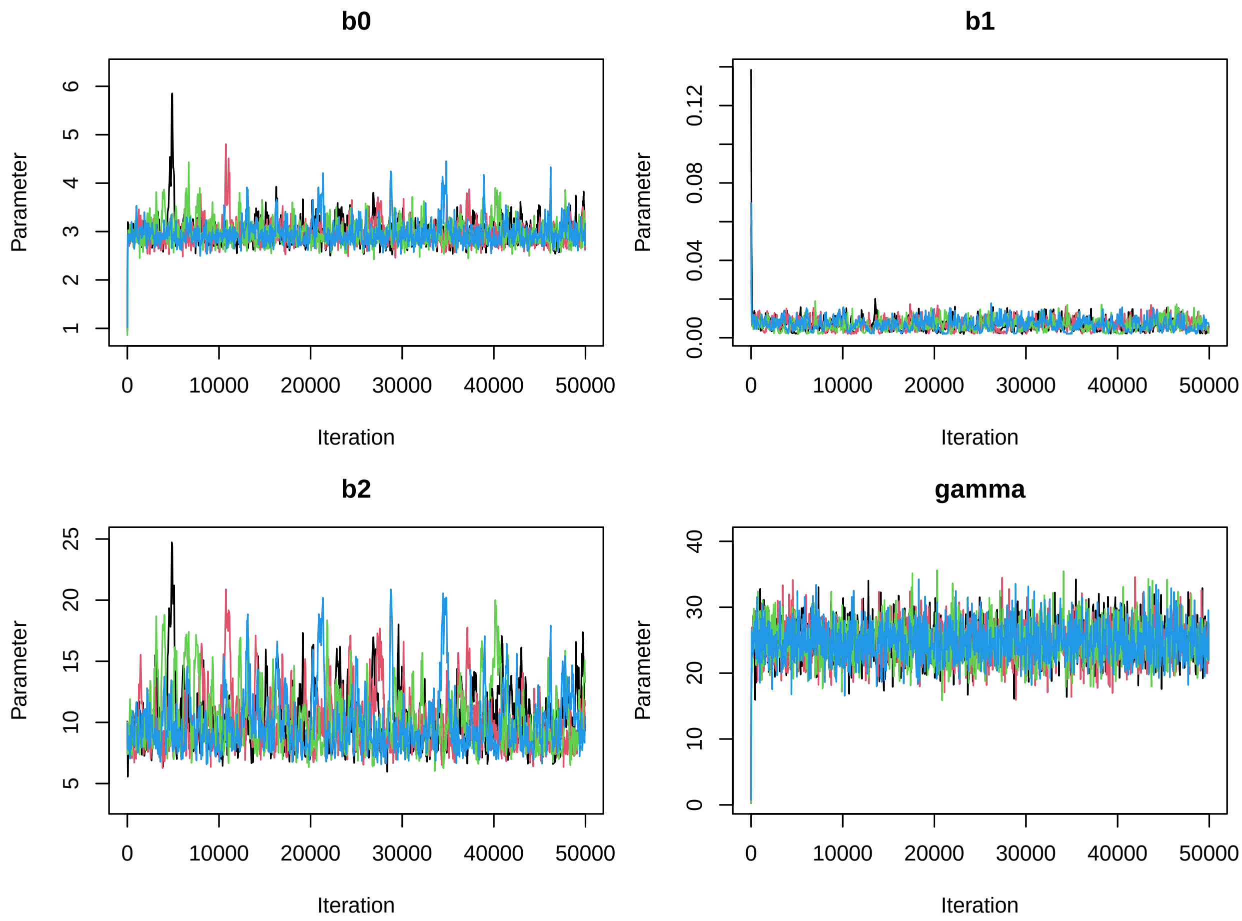
```r
# Save output
saveRDS(out_quad_OR, here("bafta_outputs", "out_quad_OR.rds"))
```

```r
# Load saved model
out_quad_OR <- readRDS(here("bafta_outputs", "out_quad_OR.rds"))
out_quad_OR
```

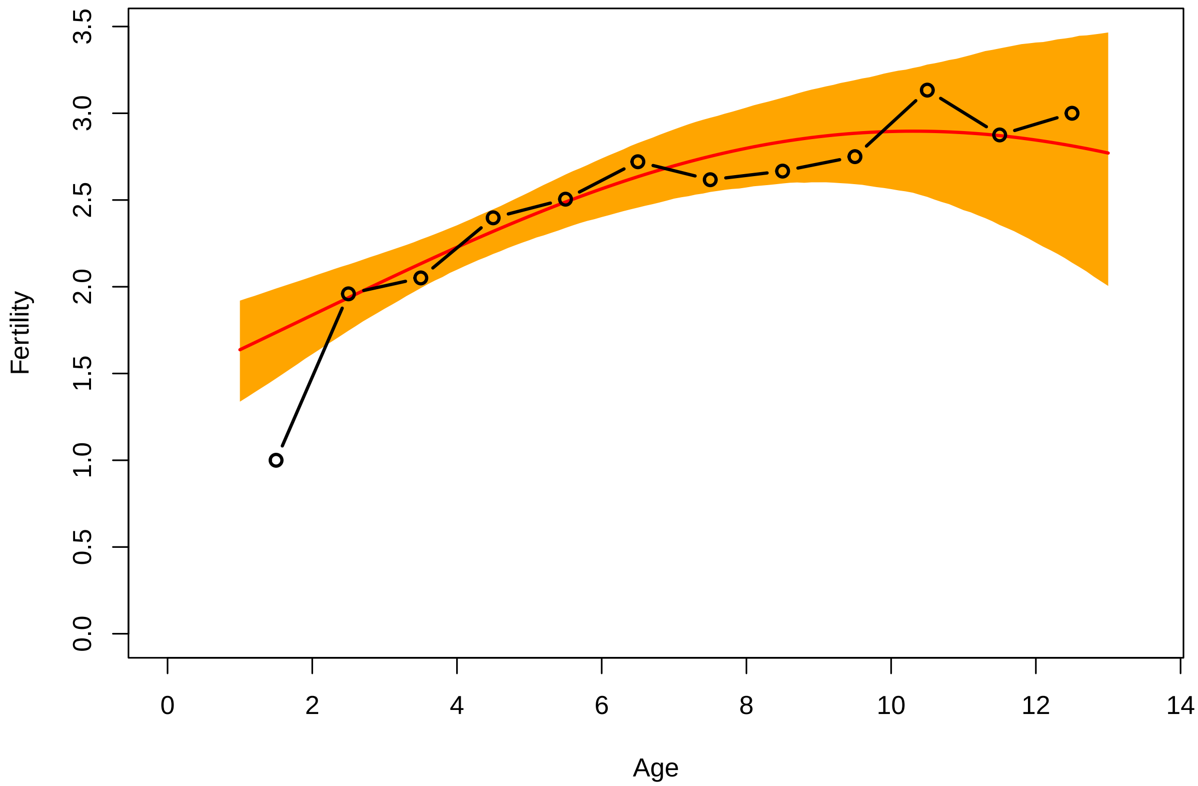
```
##
## Call:
## Model             : quadratic
## Data type         : indivSimple
## Num. iterations   : 50000
## Burnin            : 10000
## Thinning          : 50
## Number of sims.   : 4
## Computing time    : 1.49 mins
##
## Coefficients:
##            Mean       SD     Lower   Upper Neff Rhat
## b0     2.977221 0.249244  2.622065  3.5761  994    1
## b1     0.006729 0.003052  0.002288  0.0138  870    1
## b2    10.094630 2.438411  6.962768 16.2872  664    1
## gamma 24.725411 2.944696 19.213635 30.6944 3189    1
##
## Convergence:
## All parameter chains converged.
##
## Model fit:
##
## DIC = 2224.25
##
## Predictive loss:
##  Good. Fit Penalty Deviance
##        526    1866     2392
```

Results: - Convergence: All parameter chains converged - DIC = 2224.25 - Predictive loss: Good (Fit = 526, Penalty = 1866, Deviance = 2392) - Smallest Neff = 664 - Ok

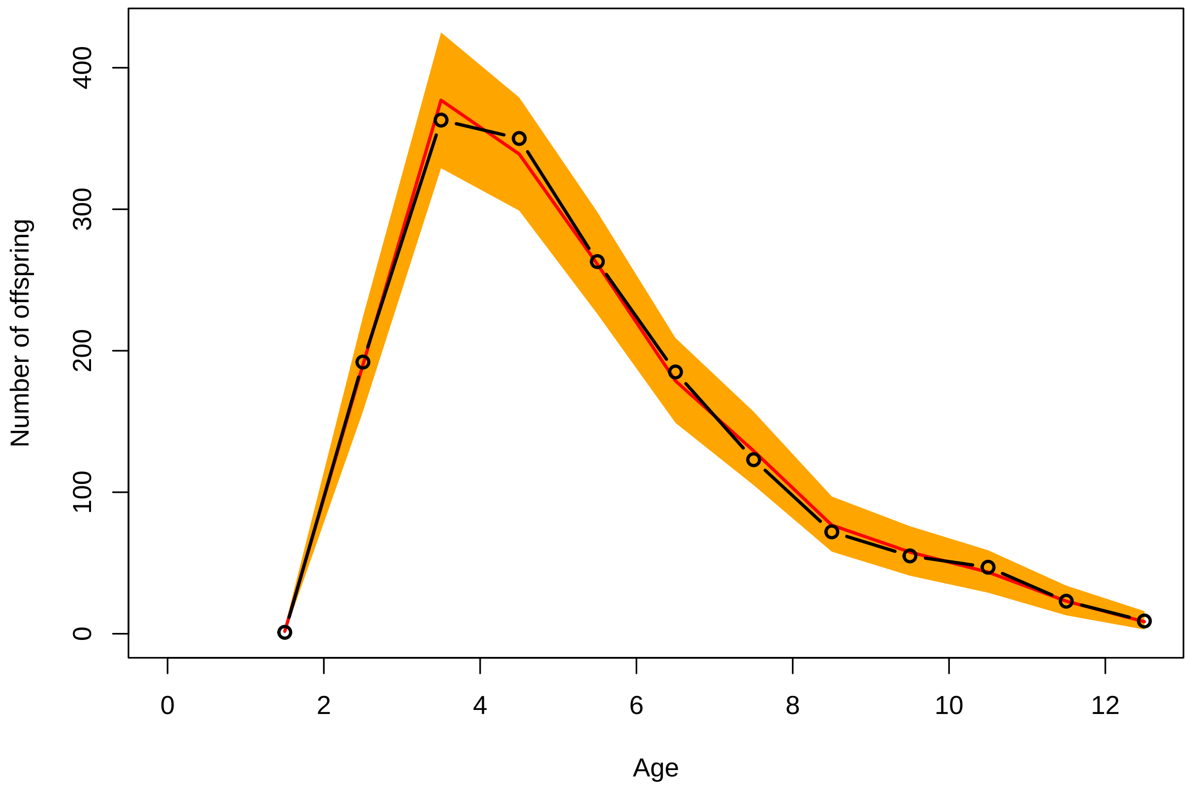
```r
# Diagnostic plots
plot(out_quad_OR)
```




```
plot(out_quad_OR, type = "fertility")
```




```
plot(out_quad_OR, type = "predictive")
```



OR: Gamma model

```
# Run model (long computation)
set.seed(123)

out_gamma_OR <- bafta(object = dataOR, model = "gamma",
                      dataType = "indivSimple", nsim = 4, ncpus = 4,
                      niter = 50000, burnin = 10000, thinning = 50)

# Save output
saveRDS(out_gamma_OR, here("bafta_outputs", "out_gamma_OR.rds"))
```

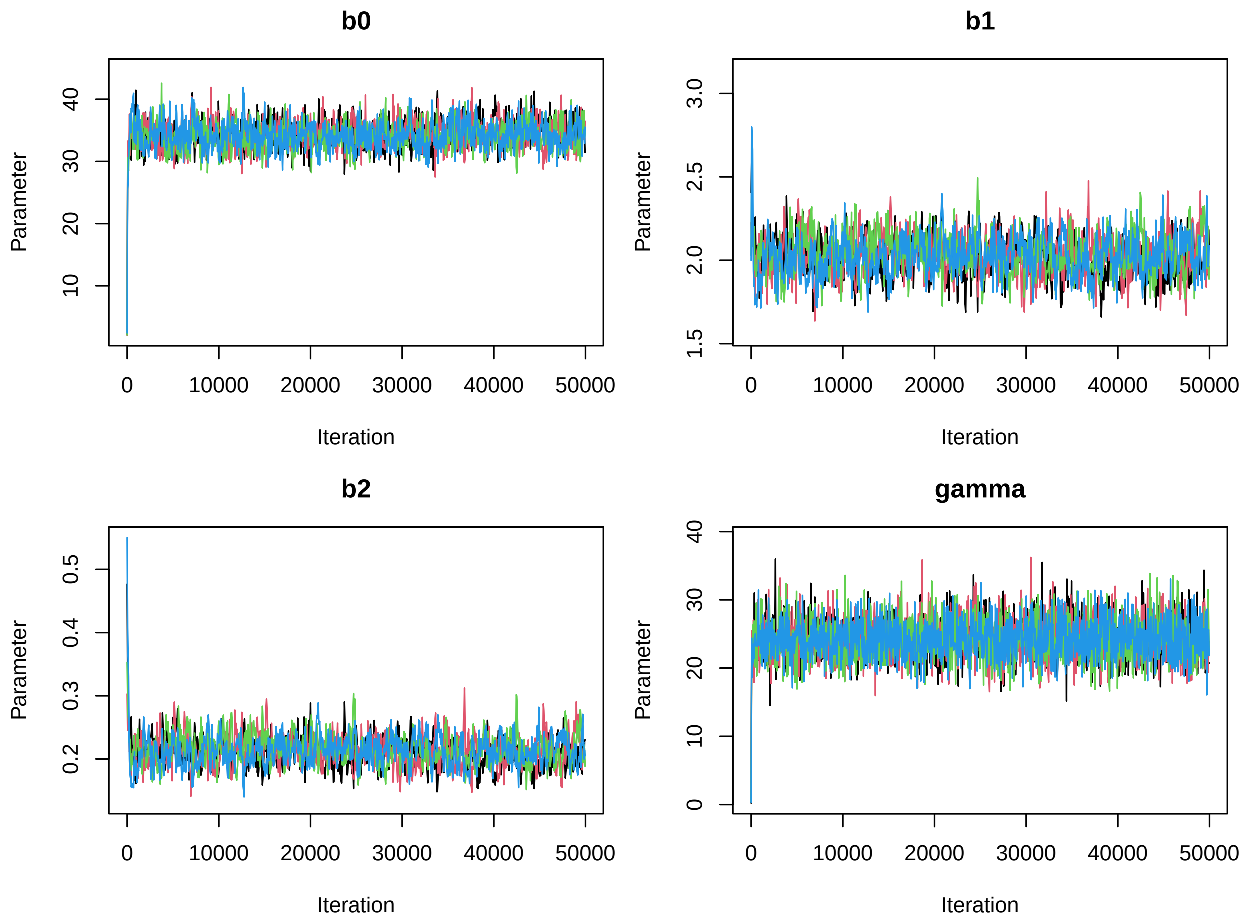
```
# Load saved model
out_gamma_OR <- readRDS(here("bafta_outputs", "out_gamma_OR.rds"))
out_gamma_OR
```

```
##
## Call:
## Model              : gamma
## Data type          : indivSimple
## Num. iterations    : 50000
```

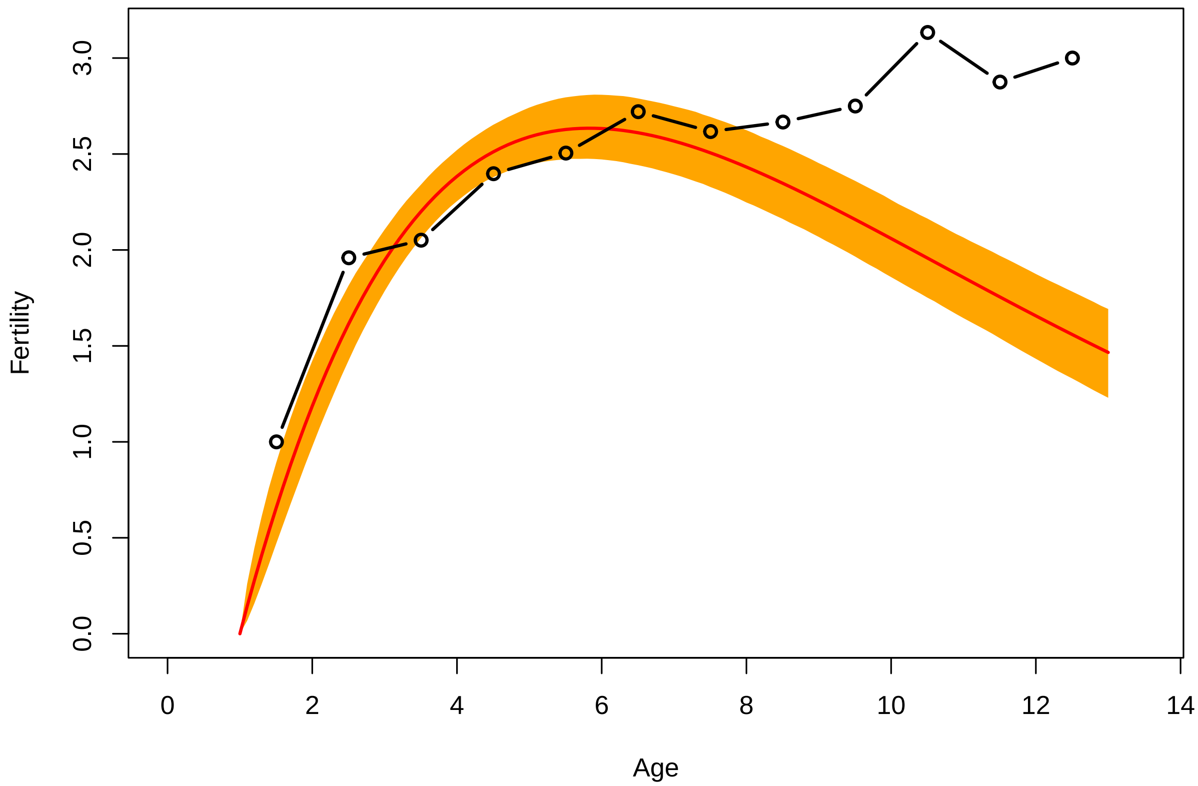
```
## Burnin           : 10000
## Thinning         : 50
## Number of sims.  : 4
## Computing time   : 3.15 mins
##
## Coefficients:
##          Mean      SD   Lower   Upper Neff  Rhat
## b0    34.3078 2.15486 30.2935 38.6388 1203 1.004
## b1     2.0255 0.11900  1.8023  2.2621  985 1.009
## b2     0.2123 0.02321  0.1703  0.2604  825 1.009
## gamma 24.3164 2.99494 18.8148 30.5560 1068 1.000
##
## Convergence:
## All parameter chains converged.
##
## Model fit:
##
## DIC = 2270.91
##
## Predictive loss:
##  Good. Fit Penalty Deviance
##        591    1810     2401
```

Results: - Convergence: All parameter chains converged - DIC = 2270.91 - Predictive loss: Good (Fit = 591, Penalty = 1810, Deviance = 2410) - Smallest Neff = 825 - Good

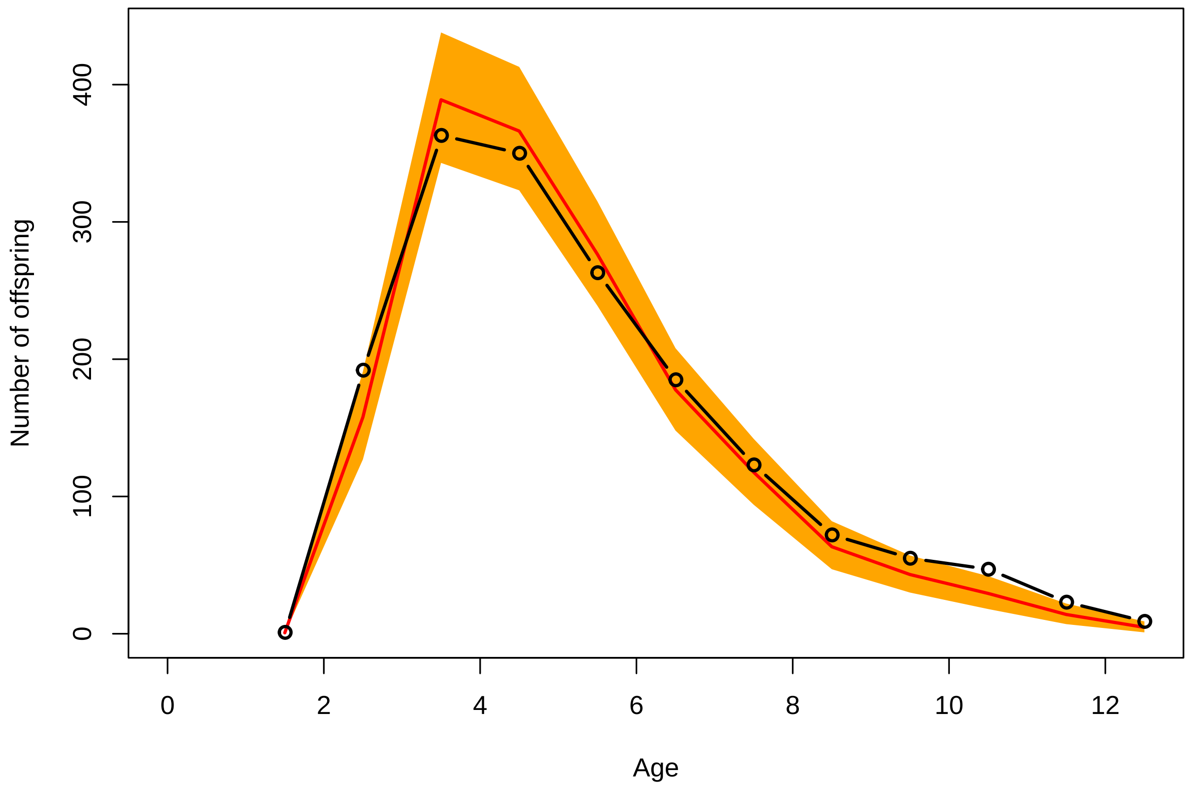
```r
# Diagnostic plots
plot(out_gamma_OR)
```




```
plot(out_gamma_OR, type = "fertility") # not good
```




```r
plot(out_gamma_OR, type = "predictive")
```



OR: Colchero-Muller model

```
# Run model (long computation)
set.seed(123)

out_CM_OR <- bafta(object = dataOR, model = "ColcheroMuller",
                   dataType = "indivSimple", nsim = 4, ncpus = 4,
                   niter = 50000, burnin = 10000, thinning = 50)

# Save output
saveRDS(out_CM_OR, here("bafta_outputs", "out_CM_OR.rds"))
```

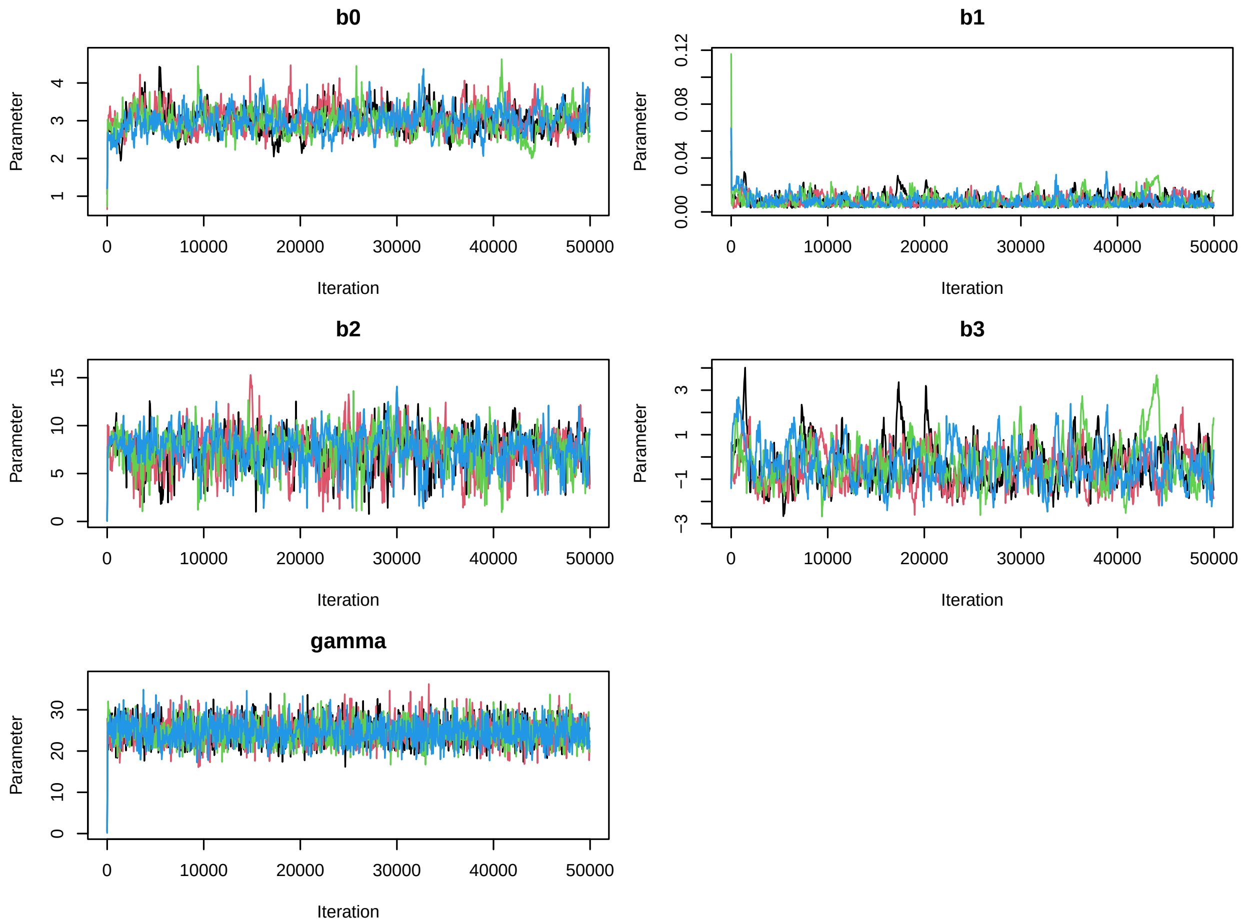
```
# Load saved model
out_CM_OR <- readRDS(here("bafta_outputs", "out_CM_OR.rds"))
out_CM_OR
```

```
##
## Call:
## Model              : ColcheroMuller
## Data type          : indivSimple
## Num. iterations    : 50000
```

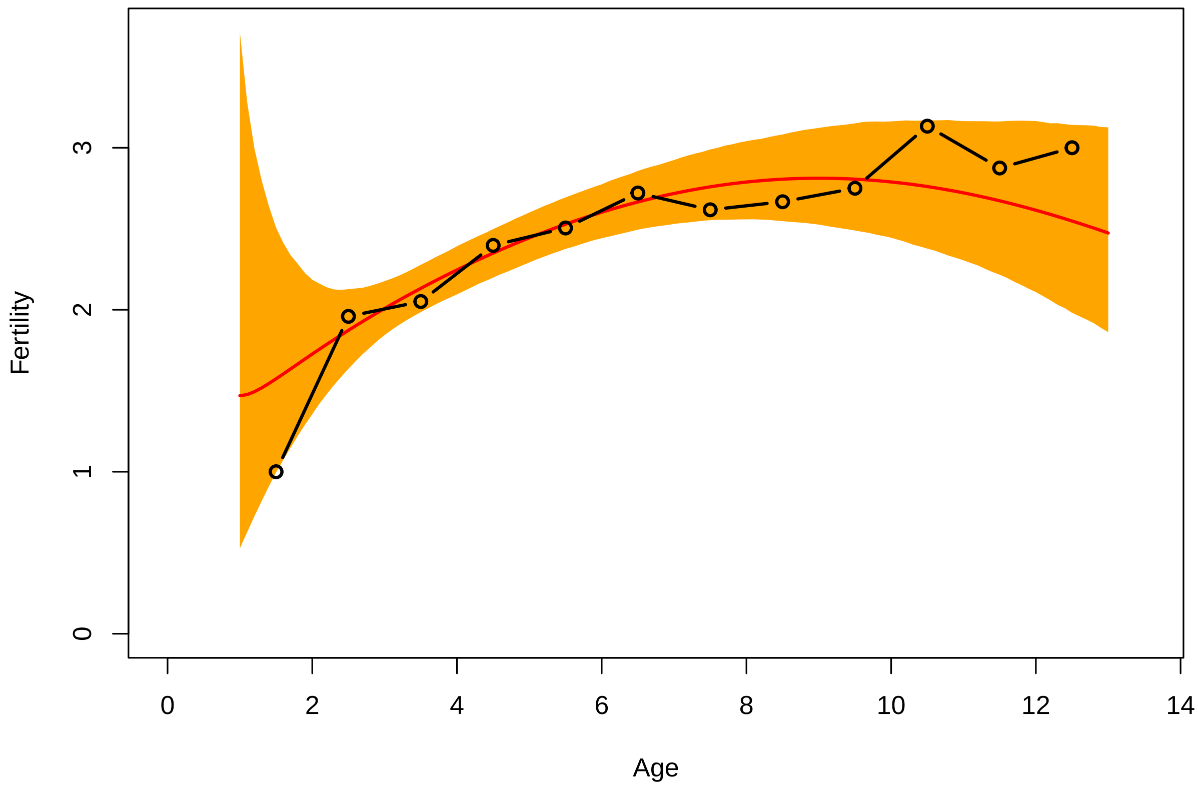
```
## Burnin           : 10000
## Thinning         : 50
## Number of sims.  : 4
## Computing time   : 2.02 mins
##
## Coefficients:
##            Mean      SD     Lower    Upper Neff  Rhat
## b0     3.003253 0.32766  2.400120  3.70422  812 1.008
## b1     0.007974 0.00414  0.003308  0.01886  763 1.008
## b2     7.466654 1.94796  2.991168 10.99635  986 1.001
## b3    -0.352923 0.90808 -1.854008  1.66477  549 1.009
## gamma 24.740458 2.94966 19.261794 30.74549 1068 1.000
##
## Convergence:
## All parameter chains converged.
##
## Model fit:
##
## DIC = 2226.68
##
## Predictive loss:
##  Good. Fit Penalty Deviance
##        529    1864     2393
```

Results: - Convergence: All parameter chains converged - DIC = 2226.68 - Predictive loss: Good (Fit = 529, Penalty = 1864, Deviance = 2393) - Smallest Neff = 549 - OK

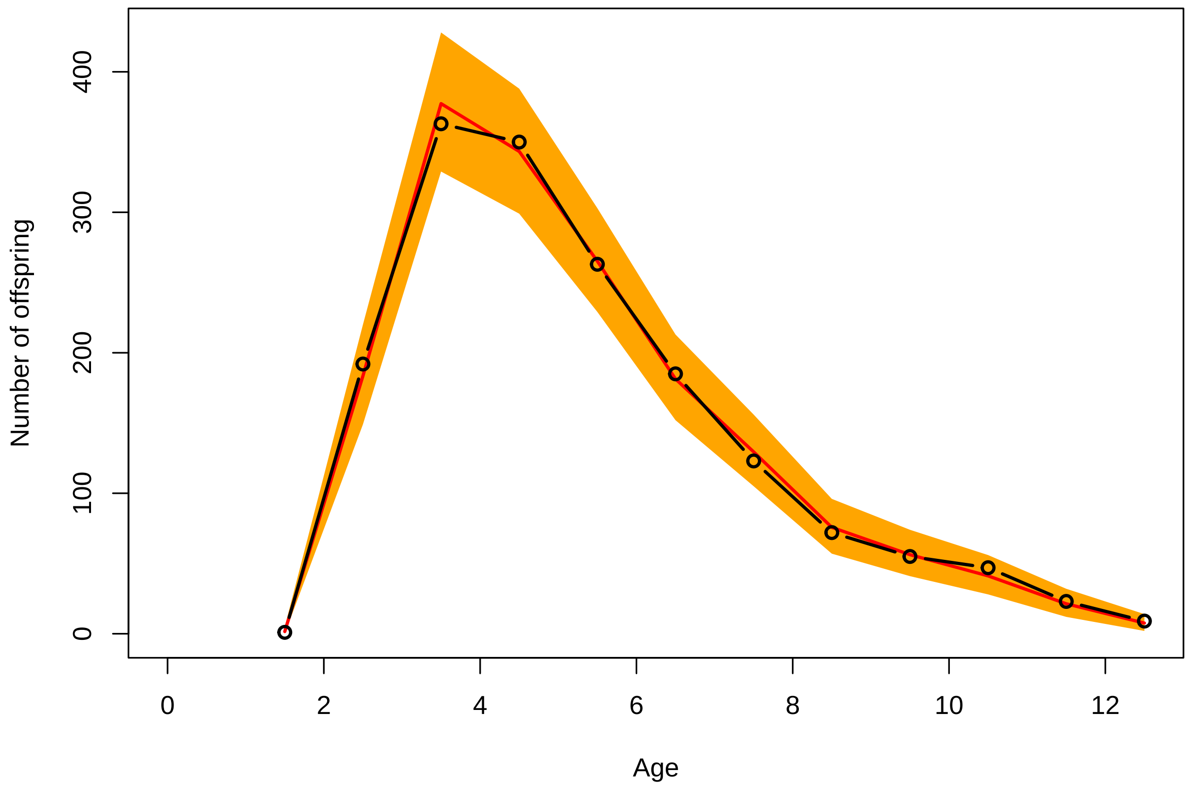
```
# Diagnostic plots
plot(out_CM_OR)
```




```r
plot(out_CM_OR, type = "fertility")
```




```r
plot(out_CM_OR, type = "predictive")
```



OR Model comparison

Best model for OR: CM (DIC = 2226.68) - Essentially the same as quadratic DIC is 2.43 away - Visually quadratic is the best fit, but CM is essentially equal - just poor fit at ages with low data - CM model preferred for direct comparison with CP

Final Model Selection

Decision: Use **Colchero-Muller model for both populations** to allow direct comparison of parameters across populations.

BaFTA results support previous modeling: - No real difference in litter size trajectories between sites - No evidence of senescence, rather increase and plateau pattern

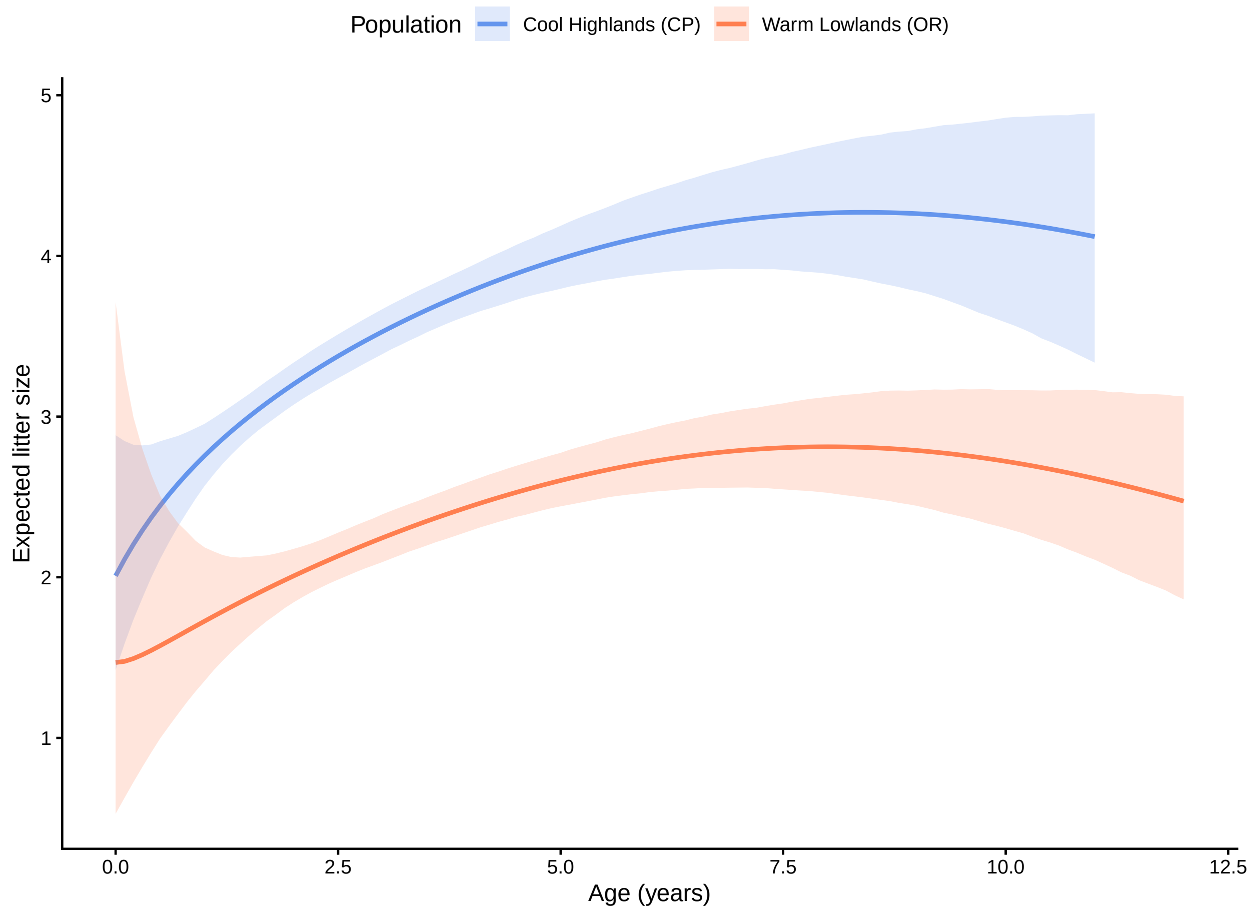
Extract Fertility Predictions

```r
# CP population
fert_CP <- data.frame(
  Age = out_CM_CP$x,
  Mean = out_CM_CP$fert[, "Mean"],
  Lower = out_CM_CP$fert[, "Lower"],
  Upper = out_CM_CP$fert[, "Upper"],
  Population = "Cool Highlands (CP)"
)

# OR population
fert_OR <- data.frame(
  Age = out_CM_OR$x,
  Mean = out_CM_OR$fert[, "Mean"],
  Lower = out_CM_OR$fert[, "Lower"],
  Upper = out_CM_OR$fert[, "Upper"],
  Population = "Warm Lowlands (OR)"
)

# Combine
fert_combined <- rbind(fert_CP, fert_OR)
```

Visualization

Fertility trajectories only

```r
ggplot(fert_combined, aes(x = Age, y = Mean, color = Population, fill = Population)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = Lower, ymax = Upper), alpha = 0.2, color = NA) +
  scale_color_manual(values = c("Cool Highlands (CP)" = "cornflowerblue",
                               "Warm Lowlands (OR)" = "coral")) +
  scale_fill_manual(values = c("Cool Highlands (CP)" = "cornflowerblue",
                              "Warm Lowlands (OR)" = "coral")) +
  labs(x = "Age (years)",
       y = "Expected litter size",
       color = "Population",
       fill = "Population") +
  theme_classic() +
  theme(legend.position = "top")
```

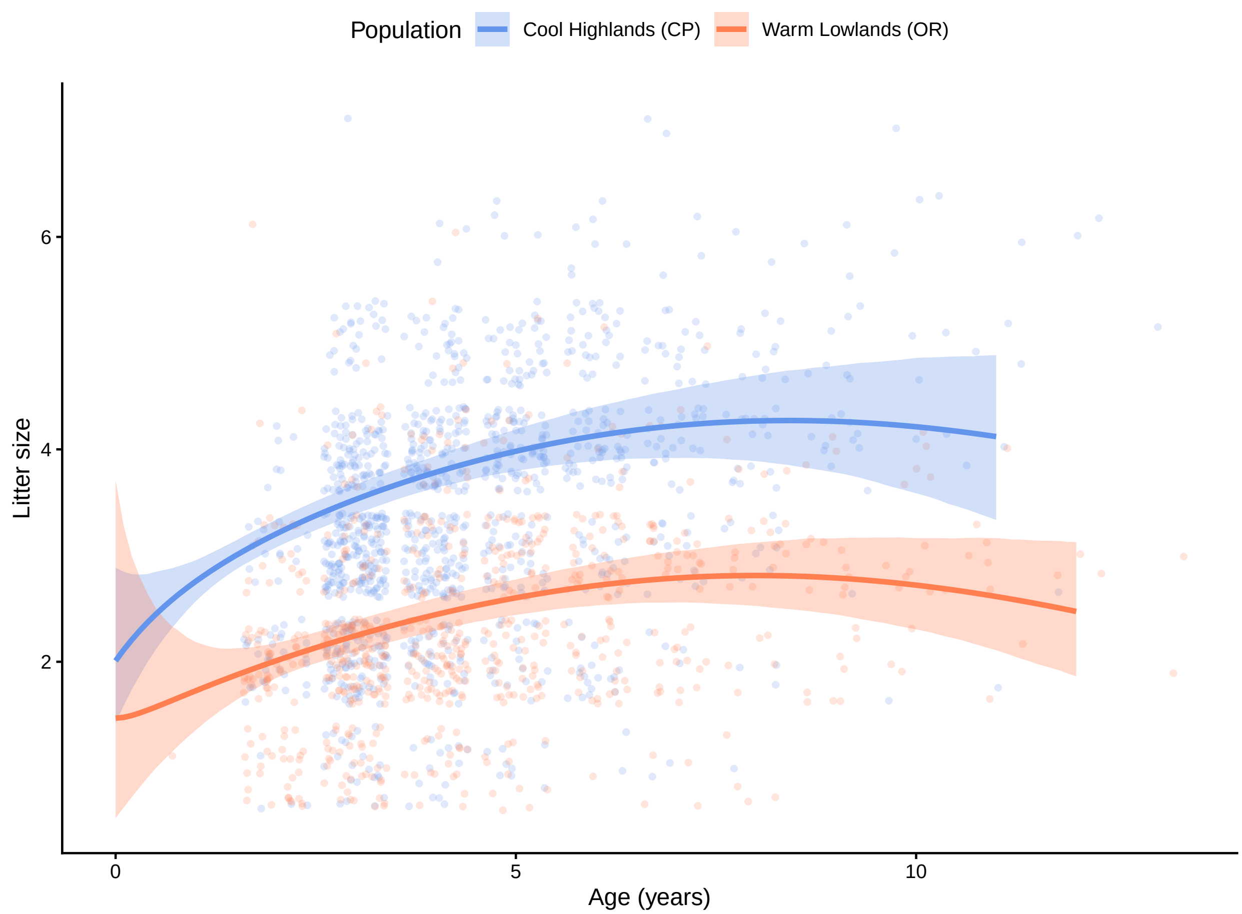



```
# Save
ggsave(here("outputs", "figures", "fertility_comparison.pdf"), width = 7, height = 5)
ggsave(here("outputs", "figures", "fertility_comparison.png"), width = 7, height = 5)
```

Fertility trajectories with raw data

```
ggplot() +
  # Raw data points (semi-transparent)
  geom_jitter(data = dataCP, aes(x = Age, y = nOffspring),
              color = "cornflowerblue", alpha = 0.2, size = 1) +
  geom_jitter(data = dataOR, aes(x = Age, y = nOffspring),
              color = "coral", alpha = 0.2, size = 1) +
  # Fitted curves with confidence intervals
  geom_ribbon(data = fert_combined,
               aes(x = Age, ymin = Lower, ymax = Upper, fill = Population),
               alpha = 0.3) +
  geom_line(data = fert_combined,
             aes(x = Age, y = Mean, color = Population),
             linewidth = 1.2) +
  scale_color_manual(values = c("Cool Highlands (CP)" = "cornflowerblue",
                                "Warm Lowlands (OR)" = "coral")) +
```

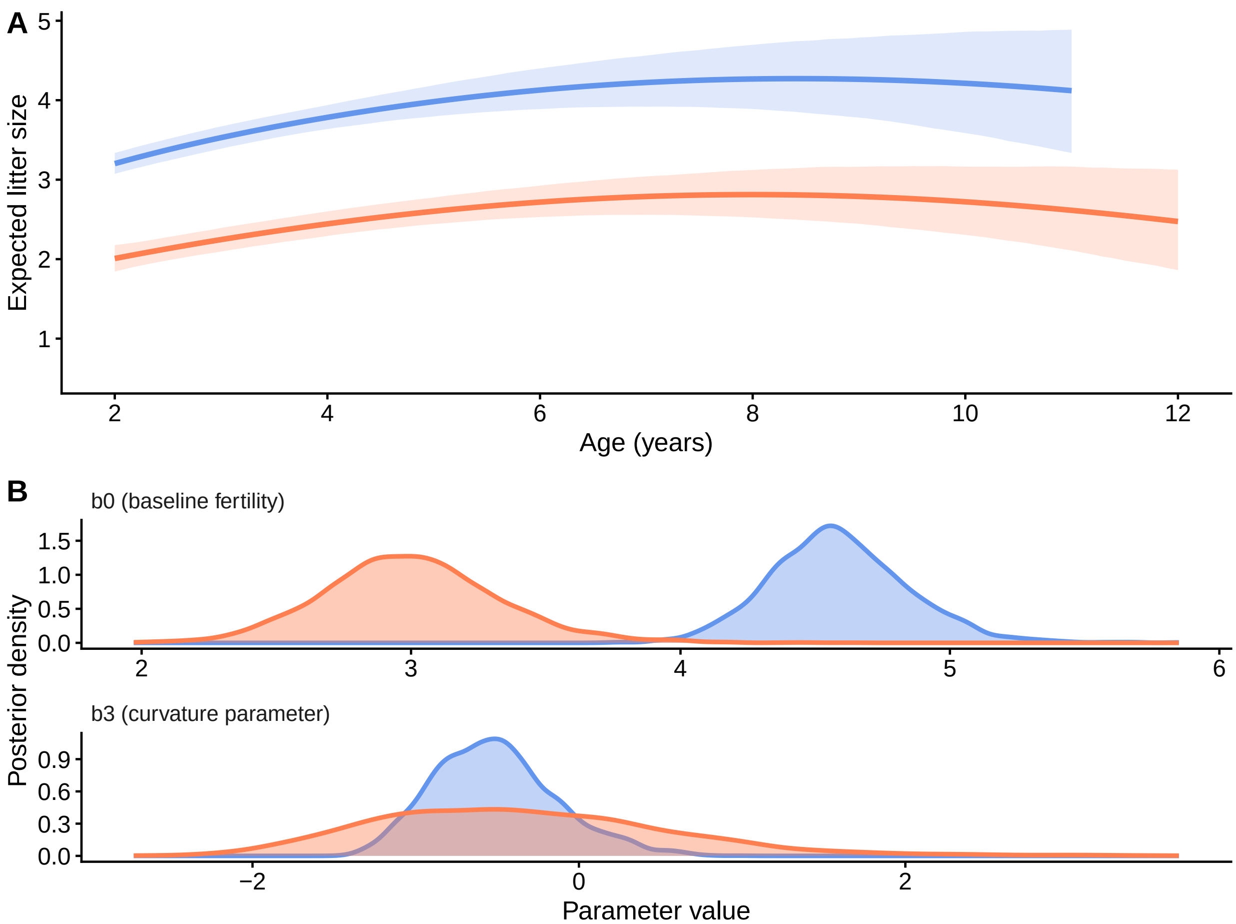
```r
  scale_fill_manual(values = c("Cool Highlands (CP)" = "cornflowerblue",
                               "Warm Lowlands (OR)" = "coral")) +
  labs(x = "Age (years)",
       y = "Litter size",
       color = "Population",
       fill = "Population") +
  theme_classic() +
  theme(legend.position = "top")
```



Fertility trajectories with parameter densities

```r
# ============== DEFINE COLOURS (single source of truth) ==============
site_colors <- c("Cool Highlands (CP)" = "cornflowerblue",
                 "Warm Lowlands (OR)"  = "coral")

# ============== EXTRACT KEY PARAMETER POSTERIORS ==============
params_OR <- as.data.frame(out_CM_OR$theta) %>%
  mutate(site = "Warm Lowlands (OR)")
params_CP <- as.data.frame(out_CM_CP$theta) %>%
  mutate(site = "Cool Highlands (CP)")
```

```r
params_combined <- bind_rows(params_OR, params_CP) %>%
  pivot_longer(cols = c(b0, b3),
               names_to = "parameter",
               values_to = "value")

# ============== PANEL A: FERTILITY TRAJECTORIES ==============
p_fertility <- ggplot(fert_combined, aes(x = Age, y = Mean,
                                         color = Population, fill = Population)) +
  geom_ribbon(aes(ymin = Lower, ymax = Upper), alpha = 0.2, color = NA) +
  geom_line(linewidth = 1.2) +
  scale_color_manual(values = site_colors) +
  scale_fill_manual(values = site_colors) +
  scale_x_continuous(breaks = seq(2, 12, 2), limits = c(2, 12)) +
  labs(y = "Expected litter size", x = "Age (years)") +
  theme_publication +
  theme(legend.position = "none")

# ============== PANEL B: KEY PARAMETER DISTRIBUTIONS ==============
p_params <- ggplot(params_combined, aes(x = value, fill = site, color = site)) +
  geom_density(alpha = 0.4, linewidth = 1) +
  facet_wrap(~parameter, scales = "free", ncol = 1,
             labeller = labeller(parameter = c(
               "b0" = "b0 (baseline fertility)",
               "b3" = "b3 (curvature parameter)"))) +
  scale_fill_manual(values = site_colors) +
  scale_color_manual(values = site_colors) +
  labs(x = "Parameter value", y = "Posterior density") +
  theme_publication +
  theme(legend.position = "none",
        strip.background = element_blank(),
        strip.text = element_text(size = 10, hjust = 0))

# ============== COMBINE PANELS ==============
combined_figure <- plot_grid(
  p_fertility, p_params,
  ncol = 1,
  labels = c("A", "B"),
  label_size = 14)

combined_figure
```




```
# ============== SAVE ==============
ggsave(here("outputs", "figures", "Figure_fertility_and_parameters.pdf"), width = 7, height = 9)
ggsave(here("outputs", "figures", "Figure_fertility_and_parameters.png"), width = 7, height = 9)
```

Extract and Export Coefficients

```
# Extract coefficient tables
coef_CP <- out_CM_CP$coefficients
coef_OR <- out_CM_OR$coefficients

# View them
print("CP coefficients:")
```

```
## [1] "CP coefficients:"
```

```
print(coef_CP)
```

```
##                Mean          SD         Lower       Upper Neff     Rhat
## b0     4.588668945 0.257074344  4.116746577  5.11210790 1007 1.004845
## b1     0.005144223 0.002911966  0.001661616  0.01234624  614 1.004782
## b2     7.916849834 2.106234512  3.664397477 12.55054026  951 1.002047
## b3    -0.513591336 0.374995466 -1.153915526  0.31765678  655 1.005427
## gamma 32.349823988 2.900752863 27.183493771 38.53556015 2738 1.000000
```

```r
print("OR coefficients:")
```

```
## [1] "OR coefficients:"
```

```r
print(coef_OR)
```

```
##                Mean          SD         Lower       Upper Neff     Rhat
## b0     3.003253343 0.327655585  2.400119854  3.70421849  812 1.008141
## b1     0.007974368 0.004140119  0.003307519  0.01885779  763 1.008207
## b2     7.466654441 1.947959420  2.991168016 10.99634769  986 1.000834
## b3    -0.352922970 0.908075237 -1.854007854  1.66476526  549 1.008701
## gamma 24.740457789 2.949664497 19.261794213 30.74549092 1068 1.000000
```

Create coefficient table for Word

```r
# Function to round to significant figures
signif_format <- function(x, digits = 3) {
  signif(x, digits)
}

# Create a combined dataframe with UNIQUE column names
coef_table <- data.frame(
  Parameter = rownames(coef_CP),
  CP_Mean = signif_format(coef_CP[, "Mean"]),
  CP_SD = signif_format(coef_CP[, "SD"]),
  CP_Lower = signif_format(coef_CP[, "Lower"]),
  CP_Upper = signif_format(coef_CP[, "Upper"]),
  OR_Mean = signif_format(coef_OR[, "Mean"]),
  OR_SD = signif_format(coef_OR[, "SD"]),
  OR_Lower = signif_format(coef_OR[, "Lower"]),
  OR_Upper = signif_format(coef_OR[, "Upper"])
)

# Create flextable
ft <- flextable(coef_table) %>%
  set_header_labels(
    Parameter = "Parameter",
    CP_Mean = "Mean",
    CP_SD = "SD",
    CP_Lower = "Lower 95% CI",
    CP_Upper = "Upper 95% CI",
    OR_Mean = "Mean",
    OR_SD = "SD",
```

```r
    OR_Lower = "Lower 95% CI",
    OR_Upper = "Upper 95% CI"
  ) %>%
  add_header_row(
    values = c("", "Cool Highlands (CP)", "Warm Lowlands (OR)"),
    colwidths = c(1, 4, 4)
  ) %>%
  align(align = "center", part = "all") %>%
  align(j = 1, align = "left", part = "body") %>%
  bold(part = "header") %>%
  fontsize(size = 10, part = "all") %>%
  font(fontname = "Times New Roman", part = "all") %>%
  border_outer(border = fp_border(width = 2), part = "all") %>%
  border_inner_h(border = fp_border(width = 1), part = "all") %>%
  border_inner_v(border = fp_border(width = 1), part = "all") %>%
  add_footer_lines("Values rounded to 3 significant figures.") %>%
  autofit()

ft
```

	Cool Highlands (CP)				Warm Lowland		
Parameter	**Mean**	**SD**	**Lower 95% CI**	**Upper 95% CI**	**Mean**	**SD**	**Lower 9**
b0	4.59000	0.25700	4.12000	5.1100	3.00000	0.32800	2.40
b1	0.00514	0.00291	0.00166	0.0123	0.00797	0.00414	0.00
b2	7.92000	2.11000	3.66000	12.6000	7.47000	1.95000	2.99
b3	-0.51400	0.37500	-1.15000	0.3180	-0.35300	0.90800	-1.85
gamma	32.30000	2.90000	27.20000	38.5000	24.70000	2.95000	19.30

Values rounded to 3 significant figures.

```r
# Save to Word document
save_as_docx(ft, path = here("outputs", "tables", "BaFTA_coefficients.docx"))

# Also print to console
print(coef_table)
```

```
##        Parameter  CP_Mean    CP_SD CP_Lower  CP_Upper  OR_Mean    OR_SD OR_Lower
## b0            b0  4.59000 0.25700  4.12000   5.1100  3.00000 0.32800  2.40000
## b1            b1  0.00514 0.00291  0.00166   0.0123  0.00797 0.00414  0.00331
## b2            b2  7.92000 2.11000  3.66000  12.6000  7.47000 1.95000  2.99000
## b3            b3 -0.51400 0.37500 -1.15000   0.3180 -0.35300 0.90800 -1.85000
## gamma      gamma 32.30000 2.90000 27.20000  38.5000 24.70000 2.95000 19.30000
##        OR_Upper
## b0       3.7000
## b1       0.0189
## b2      11.0000
## b3       1.6600
## gamma   30.7000
```

Model Comparison Table

```r
# Create model comparison table
model_comparison <- data.frame(
  Population = rep(c("Cool Highlands (CP)", "Warm Lowlands (OR)"), each = 3),
  Model = rep(c("Quadratic", "Gamma", "Colchero-Muller"), 2),
  DIC = c(out_quad_CP$DIC["DIC"], out_gamma_CP$DIC["DIC"], out_CM_CP$DIC["DIC"],
        out_quad_OR$DIC["DIC"], out_gamma_OR$DIC["DIC"], out_CM_OR$DIC["DIC"]),
  Predictive_Loss = c(out_quad_CP$PredLoss[1, "Deviance"],
                      out_gamma_CP$PredLoss[1, "Deviance"],
                      out_CM_CP$PredLoss[1, "Deviance"],
                      out_quad_OR$PredLoss[1, "Deviance"],
                      out_gamma_OR$PredLoss[1, "Deviance"],
                      out_CM_OR$PredLoss[1, "Deviance"])
)
# Display
knitr::kable(model_comparison, caption = "Model comparison for both populations")
```

Table 2: Model comparison for both populations

Population	Model	DIC	Predictive_Loss
Cool Highlands (CP)	Quadratic	4128.850	5777.100
Cool Highlands (CP)	Gamma	4237.844	5898.707
Cool Highlands (CP)	Colchero-Muller	4128.566	5772.471
Warm Lowlands (OR)	Quadratic	2224.254	2392.334
Warm Lowlands (OR)	Gamma	2270.909	2400.576
Warm Lowlands (OR)	Colchero-Muller	2226.679	2392.672

```r
# Save as supplementary table
ft_models <- flextable(model_comparison) %>%
  set_header_labels(
    Population = "Population",
    Model = "Model",
    DIC = "DIC",
    Predictive_Loss = "Predictive Loss"
  ) %>%
  bold(part = "header") %>%
  add_footer_lines("CP quadratic DIC should be interpreted with caution due to poor convergence.") %>%
  autofit()

save_as_docx(ft_models, path = here("outputs", "tables", "Supplementary_Table_ModelComparison.docx"))

ft_models
```

Population	**Model**	**DIC**	**Predictive Loss**
Cool Highlands (CP)	Quadratic	4,128.850	5,777.100
Cool Highlands (CP)	Gamma	4,237.844	5,898.707
Cool Highlands (CP)	Colchero-Muller	4,128.566	5,772.471

Population	Model	DIC	Predictive Loss
Warm Lowlands (OR)	Quadratic	2,224.254	2,392.334
Warm Lowlands (OR)	Gamma	2,270.909	2,400.576
Warm Lowlands (OR)	Colchero-Muller	2,226.679	2,392.672

CP quadratic DIC should be interpreted with caution due to poor convergence.

Lifetime sucess LM

```
# Load data
data1 <- read.csv(here("data", "processed", "reproductive_data.csv"),
                  na = c("", "NA", ".", "?"))[, c('Year', 'FemID', 'Site', 'Age', 'Clutch')]

# Rename columns
data1 <- data1 %>%
  rename(
    year = Year,
    indID = FemID,
    site = Site,
    nOffspring = Clutch)

# Format variables
data1$indID <- as.factor(data1$indID)
data1$site <- as.factor(data1$site)
data1$Age <- as.integer(data1$Age)
data1$nOffspring <- as.integer(data1$nOffspring)

# Remove missing data
data1 <- subset(data1,
             !is.na(indID) & indID != 0 &
             !is.na(Age) & Age != 0 &
             !is.na(nOffspring) & nOffspring != 0)

# Calculate lifespan
data1 <- data1 %>%
  group_by(indID) %>%
  mutate(Lifespan = max(Age, na.rm = TRUE)) %>%
  ungroup()

# Assign cohort
data1$cohort <- (data1$year - data1$Age)
data1$cohort <- as.integer(data1$year - data1$Age)


# Get total reproduction AND cohort for each female
totrepro_full <- data1 %>%
  group_by(indID) %>%
  summarise(
    total_offspring = sum(nOffspring),
    cohort = first(cohort),
```

```r
    site = first(site))

# Fit linear mixed model
totalrepro <- lmer(total_offspring ~ site + (1|cohort), data = totrepro_full)

# View results
summary(totalrepro)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: total_offspring ~ site + (1 | cohort)
##    Data: totrepro_full
##
## REML criterion at convergence: 5068.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.4157 -0.6671 -0.3111  0.4261  7.0642
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  cohort   (Intercept)  2.702   1.644
##  Residual             38.646   6.217
## Number of obs: 777, groups:  cohort, 23
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)   8.2535     0.4471  22.7786  18.460 3.38e-15 ***
## siteOR       -3.0244     0.5110 751.3988  -5.919 4.94e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##        (Intr)
## siteOR -0.357
```

```r
anova(totalrepro)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##      Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## site 1353.7  1353.7     1 751.4  35.029 4.937e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emmeans(totalrepro, ~ site)
```

```
##  site emmean    SE   df lower.CL upper.CL
##  CP     8.25 0.448 27.4     7.33     9.17
##  OR     5.23 0.548 56.0     4.13     6.33
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```